AnyCam 3.8.1 Audit Report

Release audit - 15 October 2026 - pre-warm + connection gate + B61 - tag 3.8.1

CrystalHeeler reported on 2026-10-13 and 2026-10-15 that cards and Enhanced View take 5-10 s to open, and that a card takes about 19 s to come back after Enhanced View. Scope, agreed 2026-10-15: pre-warm through go2rtc's own preload for the main and sub stream of every camera (C26), a connection gate for rate-limited cameras, and B61. Sub-stream-first is not in scope.

Design

Part	What it does
Pre-warm (C26)	Every 30 s AnyCam makes go2rtc's preloads match the cameras: the main stream and the card stream of each RTSP camera (PUT /api/preload). go2rtc's preload consumer only counts bytes, so nothing is decoded. Not preloaded: cameras that need a password, information and appliance cards, stuck cameras (B6), and a camera AnyCam has seen resetting connections (ACD) that is not behind the gate.
Watchdog	A preloaded stream whose received bytes do not grow for 2 passes (about 60 s) has its camera's preloads deleted, so go2rtc stops redialing it; it is tried again after 10 minutes.
Connection gate	go2rtc redials a dropped source after 1 s, five times, with no setting (internal/streams/producer.go, unchanged in go2rtc's main branch). For brands with throttle_type rate_limit_per_ip_tcp (Hipcam/Microseven, Sricam, Vstarcam, Wansview, Tenvis), go2rtc's source is anycam_gate.py: it takes the camera's turn from a record under a file lock, sleeps until then, and replaces itself with go2rtc's own ffmpeg copy command (os.execvp). AnyCam's own openings take turns from the same record, and an ACD escalation (30 s) is written into it.
B61	The card's ffmpeg reads go2rtc's copy; for a gated camera that open goes to 127.0.0.1, so the 5 s brand wait and back-off floor no longer apply to it. The live view proxy's own 5 s wait is removed: the gate waits at the real open.
Setting	prewarm (default on), in config.yaml, run.sh and translations/en.yaml.

Tests run

Suite	Result	Scope
Release gate	**PASS**	Eight gates. Contracts moved: the brand cooldown before go2rtc dials a rate-limited camera is now pinned on the go2rtc source and the gate (_go2rtc_camera_src, _go2rtc_gated_src, gate_main), not on the proxy.
Python integration	**PASS**	711 of 711. New section AQ: the gate's turns and escalation, the gated source split as go2rtc 1.9.14 splits it, the shared record, which streams are preloaded, passes against a fake go2rtc (add, stall, pause, return, remove), B61 in the snapshot loop, the setting. AM1 updated for the gated source.
JavaScript behaviour	**PASS**	193 of 193.
Gate as a process	**PASS**	Run as a real process on the build PC with a harmless command: the second run waited its turn (1.55 s against 0.12 s), and the command it was given ran.

Suite	Result	Scope
Field	NOTED	Not run yet; see the next section.

Not tested

- **go2rtc running the gate on the Pi.** No Linux system or go2rtc binary on the build PC. The source string is checked against go2rtc's own parsing rules (cut at '#', QuoteSplit, {output}), not against go2rtc itself.
- **The file lock.** fcntl exists only on Linux; on the build PC the lock is skipped and the tests run in one process.
- **Pre-warm against the real cameras and the DVR.** The DVR would hold up to 14 connections; its limit is not known.
- **The B61 time on a phone.** The 19 s gap is expected to shrink to the keyframe wait plus the ffmpeg start; to be measured in the field.

Best-practices compliance

Standard	Check applied	Result	Note
1.10 Exceptions	Narrow excepts	PASS	The pass loop catches aiohttp, timeout and JSON errors only.
1.3 Async	No blocking I/O on the loop	PASS	The gate record is read with asyncio.to_thread.
1.12 Logging	Truth	PASS	Pre-warm on, off and paused are logged per camera, with the reason.
6 Security	No new exposure	PASS	Sources still come only from AnyCam; the camera address reaches the command line as go2rtc's own ffmpeg copy already did.
Rule 9 (privacy)	New text	PASS	Made-up addresses only.

Known issues

- **Phone live view (B62):** where WebRTC does not connect, MSE can fall behind; under investigation.
- **Extra connections:** each camera holds one or two all the time; the prewarm setting turns them off.
- **Microseven:** a camera in a locked state needs a power cycle to recover.

Files changed

File	Change
anycam_gate.py (new)	The connection gate: shared record, escalation, go2rtc command.
anycam_go2rtc.py	Gated source, pre-warm passes and watchdog, supervisor start/stop, no wait in the proxy, _go2rtc_will_relay.

File	Change
camera_discovery.py	Turns from the gate's record; ACD escalation written to it; CFG_PREWARM; 3.8.1.
anycam_snap.py	B61: no brand wait or floor for a gated camera read through go2rtc.
config.yaml, run.sh, translations/en.yaml	The prewarm setting.
anycam_modules.py, Dockerfile	anycam_gate.py added.
verify_release.py	Contracts moved to the go2rtc source and the gate.
tests/	Section AQ; AM1 updated.
CHANGELOG.md, CLAUDE.md, docs	3.8.1; build plan.